

Data modeling: CSCI E-106

Building the Regression Model II: Diagnostic

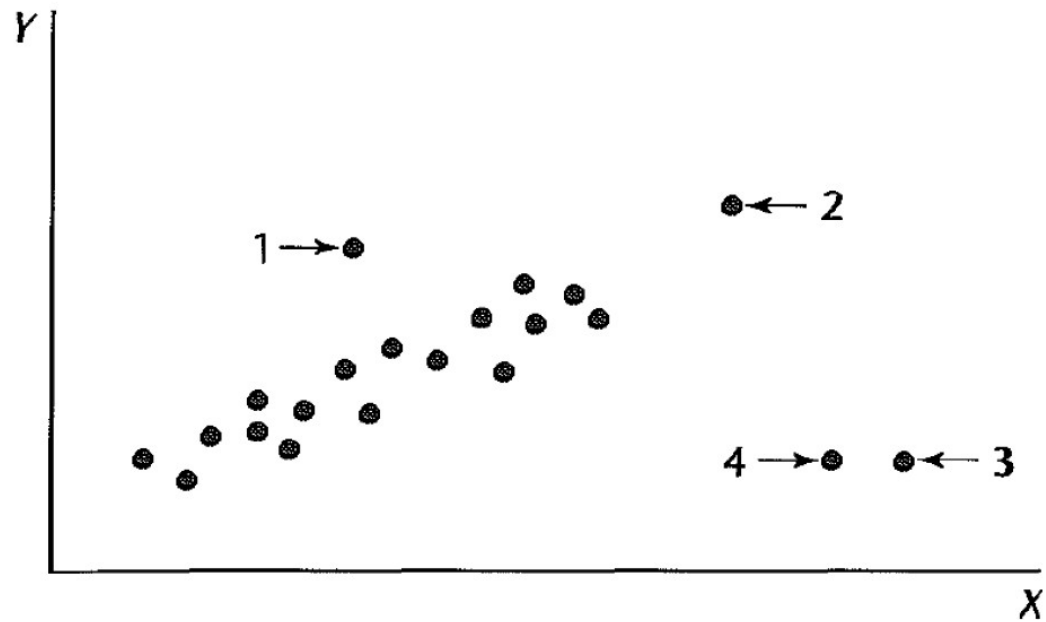
Overview

- a number of refined diagnostics for checking the adequacy of a regression model
 - detecting improper functional form for a predictor variable
 - outliers
 - influential observations
 - multicollinearity

Identifying Outlying Y Observations

Outlying or Extreme:

- the observations for these cases are well separated from the remainder of the data
- large residuals; have dramatic effects



Identifying Outlying Y Observations, cont'd

- Outlying: Y value, X values or both
- Not all outlying cases have a strong influence on the fitted regression function.
- A basic step: determine if the regression model under consideration is heavily influenced by one or a few cases in the data set

Identifying Outlying Y Observations, cont'd

Two refined measures for identifying cases with **outlying Y observations**:

- Residuals, Semistudentized Residuals:

$$e_i = Y_i - \hat{Y}_i; \quad e_i^* = \frac{e_i}{\sqrt{MSE}}$$

- Hat matrix: **$H = X(X'X)^{-1}X' \Rightarrow \hat{Y} = HY$**

$$\mathbf{e} = (\mathbf{I} - \mathbf{H})\mathbf{Y}$$

$$\Rightarrow \sigma^2\{\mathbf{e}\} = \sigma^2(\mathbf{I} - \mathbf{H})$$

$$\sigma^2\{e_i\} = \sigma^2(1 - h_{ii}), \quad h_{ii}: \text{the } i\text{th elements on the diagonal of } \mathbf{H}$$

$$\sigma\{e_i, e_j\} = -h_{ij}\sigma^2, \quad i \neq j$$

$$\Rightarrow s^2\{e_i\} = MSE(1 - h_{ii}), \quad s\{e_i, e_j\} = -h_{ij}MSE$$

Identifying Outlying Y Observations, cont'd

- $h_{ii} = \mathbf{X}_i'(\mathbf{X}'\mathbf{X})^{-1}\mathbf{X}_i$, $\mathbf{X}_i = [1 \ X_{i,1} \ \cdots \ X_{i,p-1}]'$

Small data set: $n = 4$

(a) Data and Basic Results							
i	(1) X_{i1}	(2) X_{i2}	(3) Y_i	(4) $\hat{Y}_i$	(5) e_i	(6) h_{ii}	(7) $s^2\{e_i\}$
1	14	25	301	282.2	18.8	.3877	352.0
2	19	32	327	332.3	-5.3	.9513	28.0
3	12	22	246	260.0	-14.0	.6614	194.6
4	11	15	187	186.5	.5	.9996	.2

(b) H				(c) $s^2\{e\}$					
$\begin{bmatrix}$	.3877	.1727	.4553	-.0157	$\begin{bmatrix}$	352.0	-99.3	-261.8	9.0
$\begin{bmatrix}$	.1727	.9513	-.1284	.0044	$\begin{bmatrix}$	-99.3	28.0	73.8	-2.5
$\begin{bmatrix}$	.4553	-.1284	.6614	.0117	$\begin{bmatrix}$	-261.8	73.8	194.6	-6.7
$\begin{bmatrix}$	-.0157	.0044	.0117	.9996	$\begin{bmatrix}$	9.0	-2.5	-6.7	.2

- $\hat{Y} = 80.93 - 5.84X_1 + 11.32X_2$
- $MSE = 574.9$
- $s^2\{e_1\} = 574.9(1 - 0.3877) = 352.0$

Identifying Outlying Y Observations, cont'd

- Deleted Residuals: The difference between Y_i and $\hat{Y}_{i(i)}$: (*PRESS prediction error*)

$$\text{deleted residual: } d_i = Y_i - \hat{Y}_{i(i)} = \frac{e_i}{1 - h_{ii}}$$

- h_{ii} : the larger will be the deleted residuals as compared to the ordinary residual
- the estimated variance of d_i :

$$s^2\{d_i\} = MSE_{(i)}(1 + \mathbf{X}'_i(\mathbf{X}'_{(i)}\mathbf{X}_{(i)})^{-1}\mathbf{X}_i) = \frac{MSE_{(i)}}{1 - h_{ii}}$$

$$\Rightarrow \frac{d_i}{s\{d_i\}} \sim t((n - 1) - p)$$

Identifying Outlying Y Observations, cont'd

- Studentized Deleted Residuals:

$$t_i = \frac{d_i}{s\{d_i\}} = \frac{e_i}{\sqrt{MSE_{(i)}(1 - h_{ii})}}$$
$$\left((n - p)MSE = (n - p - 1)MSE_{(i)} + \frac{e_i^2}{1 - h_{ii}} \right)$$
$$\Rightarrow t_i = e_i \left[\frac{n - p - 1}{SSE(1 - h_{ii}) - e_i^2} \right]^{1/2}$$

- h_{ii} : the larger will be the deleted residuals as compared to the ordinary residual

Identifying Outlying Y Observations, cont'd

- the estimated variance of d_i :

$$s^2\{d_i\} = MSE_{(i)}(1 + \mathbf{x}'_i(\mathbf{x}'_{(i)}\mathbf{x}_{(i)})^{-1}\mathbf{x}_i) = \frac{MSE_{(i)}}{1 - h_{ii}}$$

$$\Rightarrow \frac{d_i}{s\{d_i\}} \sim t((n - 1) - p)$$

- **Test for Outliers:** whose studentized deleted residuals are **large in absolute value**
 - If the regression model is appropriate, so that no case is outlying. Each $t_i \sim t(n - p - 1)$.
 - $|t_i|$: the appropriate Bonferroni critical value:
 $t(1 - \alpha/2n; n - p - 1)$

Identifying Outlying Y Observations, cont'd

Body fat example

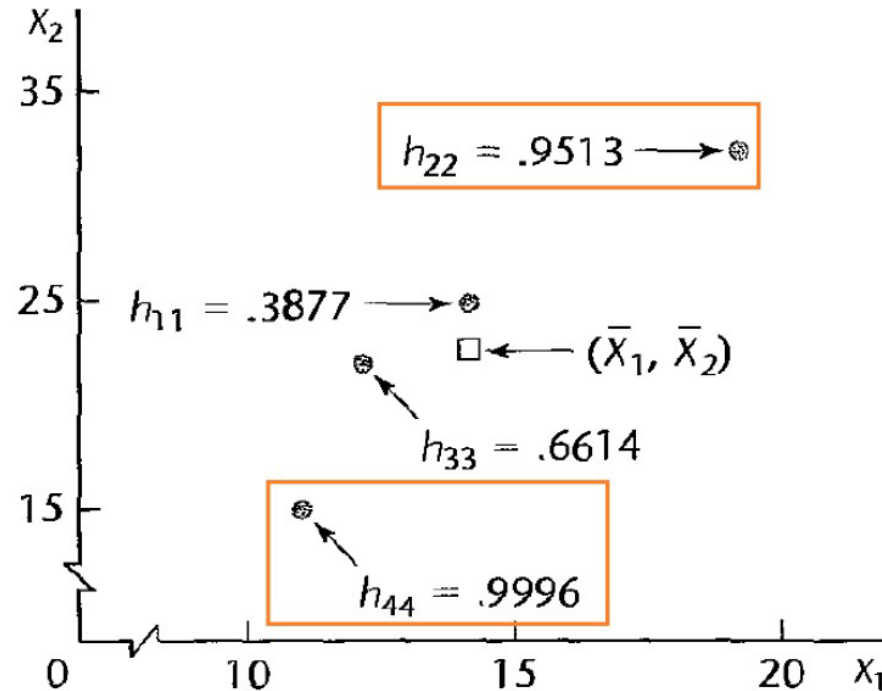
i	(1) e_i	(2) h_{ii}	(3) t_i
1	-1.683	.201	-.730
2	3.643	.059	1.534
3	-3.176	.372	<u>-1.656</u>
4	-3.158	.111	-1.348
5	.000	.248	.000
6	-.361	.129	-.148
7	.716	.156	.298
8	4.015	.096	<u>1.760</u>
9	2.655	.115	1.117
10	-2.475	.110	-1.034
11	.336	.120	.137
12	2.226	.109	.923
13	-3.947	.178	<u>-1.825</u>
14	3.447	.148	1.524
15	.571	.333	.267
16	.642	.095	.258
17	-.851	.106	.344
18	-.783	.197	.335
19	-2.857	.067	-1.176
20	1.040	.050	.409

- $|t| < 3.252 = t(1 - \alpha/2n; n - p - 1)$
- The Bonferroni procedure provides a **Conservative test** for the presence of an outlier.

Identifying Outlying X Observations

- Using **H** for identifying outlying X
- $0 \leq h_{ii} \leq 1$ $\sum_{i=1}^n h_{ii} = p$
- h_{ii} : called *leverage*; measure the distance between X_i and $\bar{X}$
 - large $h_{ii} \Rightarrow X_i$ distant from the center of all X s

FIGURE 10.6
Illustration of
Leverage
Values as
Distance
Measures—
Table 10.2
Example.



Identifying Outlying X Observations, cont'd

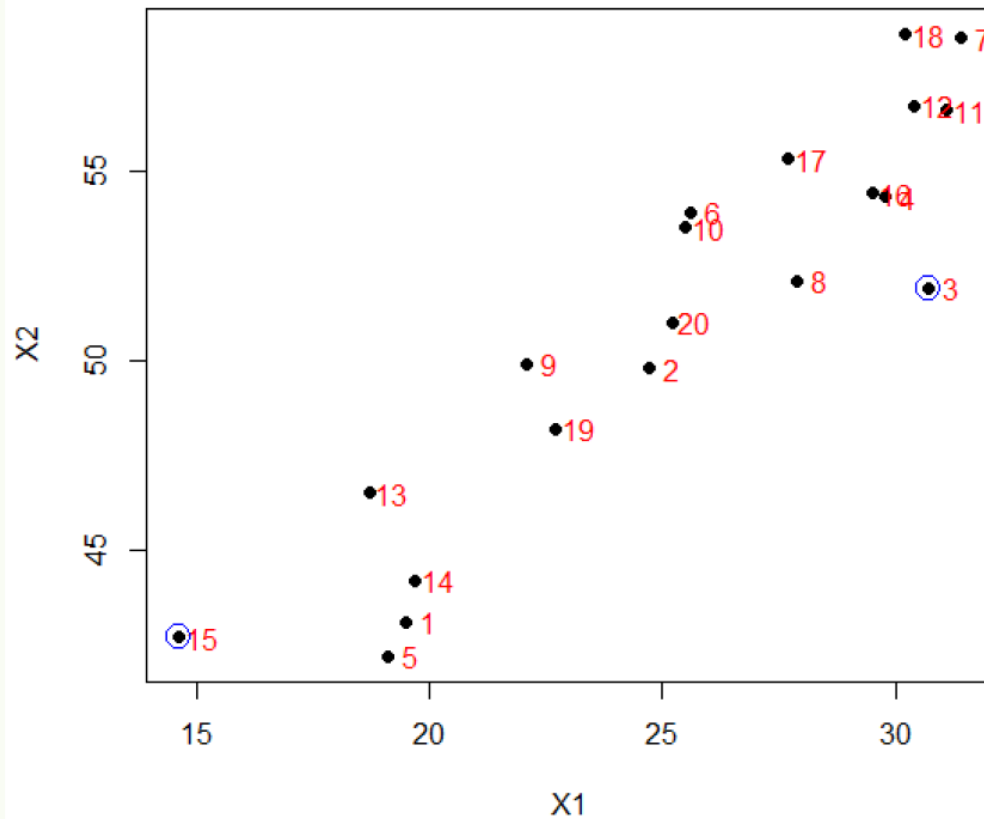
- If X_i is **outlying** $\Rightarrow$ has a **large leverage** h_{ii}
- $\hat{Y}_i$: a linear combination of Y ($\hat{\mathbf{Y}} = \mathbf{H}\mathbf{Y}$)
- h_{ii} : the weight of $Y_i \Rightarrow$ the larger is h_{ii} , the more important is Y_i in determining $\hat{Y}_i$
- The larger is h_{ii} , the smaller is $\sigma^2\{e_i\}$.
- $h_{ii} = 1 \Rightarrow \sigma^2\{e_i\} = 0$
- Rule:

$$h_{ii} > 2\bar{h} = 2\frac{\sum h_{ii}}{n} = 2\frac{p}{n} \quad \left(\frac{2p}{n} \leq 1\right)$$

$$\begin{cases} \text{very high leverage: } h_{ii} > 0.5 \\ \text{moderate leverage: } h_{ii} : 0.2 \sim 0.5 \end{cases}$$

Identifying Outlying X Observations, cont'd

Body fat example



- $2p/n = 0.30$
- $h_{33} = 0.372;$
 $h_{15,15} = 0.333$

Identifying Outlying X Observations, cont'd

```
attach(Dataset_07TA01)
fit<-lm(Y~X1+X2+X3)
n<-length(Y); p = 3
plot(X2~X1, pch=16 )
text(X1+0.5, X2,
labels=as.character(1:length(X1)),col="red")
hii<-hatvalues(fit)
index<-hii>2*p/n
points( X1[index], X2[index], cex=2.0, col="blue")
```

DFFITS measure

- Influence on **single** fitted value: *DFFITS*-measure the influence that case i has on $\hat{Y}_i$

$$(DFFITS)_i = \frac{\hat{Y}_i - \hat{Y}_{i(i)}}{\sqrt{MSE_{(i)} h_{ii}}} = t_i \left(\frac{h_{ii}}{1 - h_{ii}} \right)^{1/2}$$

- Rule:

$$\begin{cases} |DFFITS| > 1 & \text{for small to medium data sets} \\ |DFFITS| > 2\sqrt{p/n} & \text{for large data sets} \end{cases}$$

- If X_i is an outlier and has a high h_{ii} , $(DFFITS)_i$ will tend to be large absolutely.

Cook's Distance

- Influence on **all** fitted value: *Cook's distance*-consider the influence of the i th case on all n fitted values

$$D_i = \frac{\sum_{j=1}^n (\hat{Y}_i - \hat{Y}_{j(i)})^2}{pMSE} = \frac{(\hat{\mathbf{Y}} - \hat{\mathbf{Y}}_{(i)})'(\hat{\mathbf{Y}} - \hat{\mathbf{Y}}_{(i)})}{pMSE}$$
$$= \frac{e_i^2}{pMSE} \left[\frac{h_{ii}}{(1 - h_{ii})^2} \right]$$

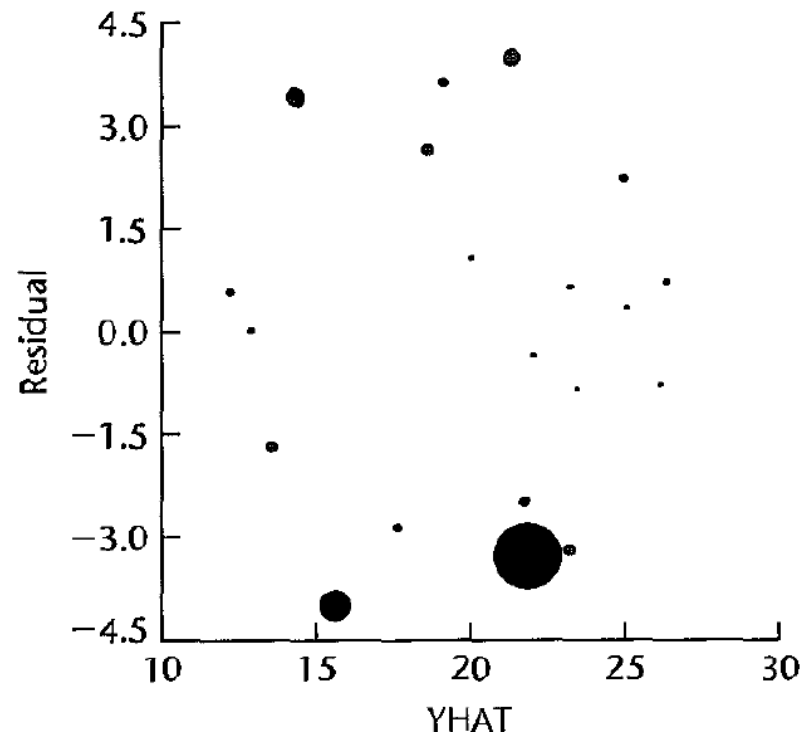
- 1 the size of e_i
- 2 the leverage h_{ii}
- 3 $e_i \uparrow$ or $h_{ii} \uparrow \Rightarrow D_i \uparrow$

- Rule: $D_i \sim F(p, n - p)$

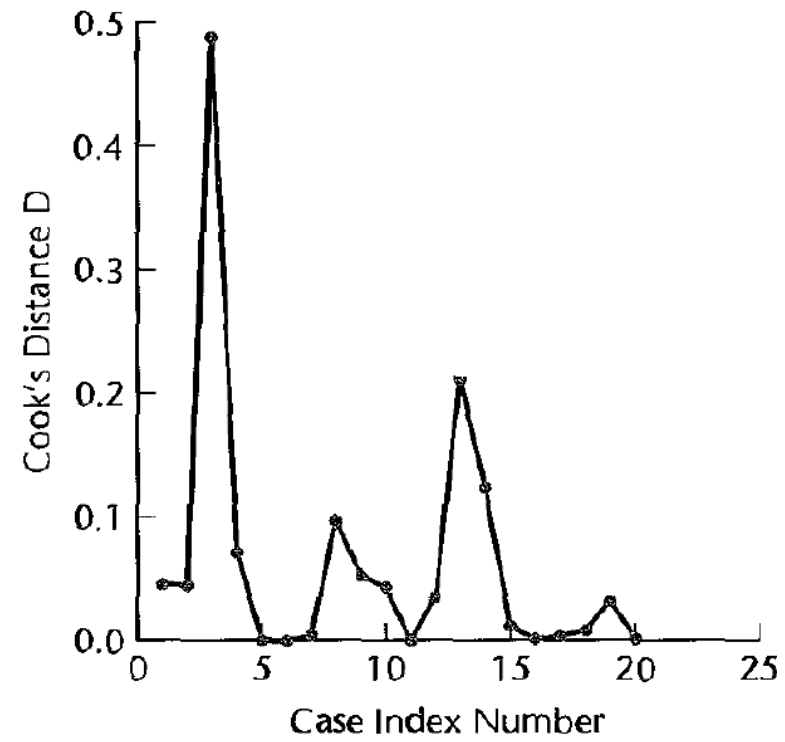
$$\begin{cases} \text{little influence : } P(F(p, n - p) \leq D_i) > 0.1 \text{ or } 0.2 \\ \text{major influence : } P(F(p, n - p) \leq D_i) > 0.5 \end{cases}$$

Cook's Distance, cont'd

(a) Proportional Influence Plot



(b) Index Influence Plot



DFBETAS

- Influence on regression coefficients: *DFBETAS*-the difference between b_k and $b_{k(i)}$

$$(DFBETAS)_{k(i)} = \frac{b_k - b_{k(i)}}{\sqrt{MSE_{(i)} c_{kk}}}, \quad k = 0, 1, \dots, p - 1$$

where c_{kk} : the k th diagonal element of $(\mathbf{X}'\mathbf{X})^{-1}$

- Rule:

$$\begin{cases} |DFBETAS| > 1 & \text{for small to medium data sets} \\ |DFBETAS| > 2\sqrt{n} & \text{for large data sets} \end{cases}$$

DFBETAS, cont'd

```
# Body fat example (Table 10.4)
> influence.measures(fit)
Influence measures of
      lm(formula = Y ~ X1 + X2) :

      dfb.1_    dfb.X1    dfb.X2    dffit cov.r    cook.d    hat inf
1  -3.05e-01 -1.31e-01  2.32e-01 -3.66e-01 1.361 4.60e-02 0.2010
2   1.73e-01  1.15e-01 -1.43e-01  3.84e-01 0.844 4.55e-02 0.0589
3  -8.47e-01 -1.18e+00  1.07e+00 -1.27e+00 1.189 4.90e-01 0.3719   *
4  -1.02e-01 -2.94e-01  1.96e-01 -4.76e-01 0.977 7.22e-02 0.1109
5  -6.37e-05 -3.05e-05  5.02e-05 -7.29e-05 1.595 1.88e-09 0.2480   *
6   3.97e-02  4.01e-02 -4.43e-02 -5.67e-02 1.371 1.14e-03 0.1286
7  -7.75e-02 -1.56e-02  5.43e-02  1.28e-01 1.397 5.76e-03 0.1555
8   2.61e-01  3.91e-01 -3.32e-01  5.75e-01 0.780 9.79e-02 0.0963
9  -1.51e-01 -2.95e-01  2.47e-01  4.02e-01 1.081 5.31e-02 0.1146
10  2.38e-01  2.45e-01 -2.69e-01 -3.64e-01 1.110 4.40e-02 0.1102
11 -9.02e-03  1.71e-02 -2.48e-03  5.05e-02 1.359 9.04e-04 0.1203
12 -1.30e-01  2.25e-02  7.00e-02  3.23e-01 1.152 3.52e-02 0.1093
13  1.19e-01  5.92e-01 -3.89e-01 -8.51e-01 0.827 2.12e-01 0.1784
14  4.52e-01  1.13e-01 -2.98e-01  6.36e-01 0.937 1.25e-01 0.1480
15 -3.00e-03 -1.25e-01  6.88e-02  1.89e-01 1.775 1.26e-02 0.3332   *
16  9.31e-03  4.31e-02 -2.51e-02  8.38e-02 1.309 2.47e-03 0.0953
17  7.95e-02  5.50e-02 -7.61e-02 -1.18e-01 1.312 4.93e-03 0.1056
18  1.32e-01  7.53e-02 -1.16e-01 -1.66e-01 1.462 9.64e-03 0.1968
19 -1.30e-01 -4.07e-03  6.44e-02 -3.15e-01 1.002 3.24e-02 0.0670
20  1.02e-02  2.29e-03 -3.31e-03  9.40e-02 1.224 3.10e-03 0.0501
```

DFBETAS, cont'd

Some final comments:

- Analysis of outlying and influential cases: a necessary component of good regression analysis
 - neither automatic nor foolproof
 - require good judgment by the analyst
- Methods described: ineffective
- **Extensions of the single-case diagnostic** procedures: computational requirements

Variance Inflation Factor-VIF

Some problems: multicollinearity

- Adding or deleting X : change the regression coefficients
- Extra sum of squares: depending upon which other X_k variables are already included in the model
- X_k highly correlated with each other $\Rightarrow s\{b_k\} \uparrow$
- the estimated regression coefficients individually may not be statistically significant

Variance Inflation Factor-VIF, cont'd

informal diagnostics for multicollinearity:

1. Large changes in b_k when X_k is added or deleted, or when an observation is altered or deleted
2. Nonsignificant results in individual tests on the regression coefficients for important predictor variables.
3. Estimated regression coefficients with an algebraic sign that is the opposite of that expected from theoretical consideration or prior experience.
4. Large r_{xx}
5. Wide confidence interval for β_k

Variance Inflation Factor-VIF, cont'd

Important limitations:

- do not provide **quantitative measurements**
- may not identify the nature of the multicollinearity
- sometimes the observed behavior may **occur without multicollinearity** being present

Variance Inflation Factor-VIF, cont'd

Variance Inflation Factor (VIF)

- a formal method: detecting multicollinearity; widely accepted
- measure how much the variances of b_k s are inflated as compared to when the predictor variables are not linearly related.
- Illustration:
- Variance-covariance matrix of $\mathbf{b}$: $\sigma^2\{\mathbf{b}\} = \sigma^2(\mathbf{X}'\mathbf{X})^{-1}$

Variance Inflation Factor-VIF, cont'd

Using the **standardized regression model**:

Variance-covariance matrix of $\mathbf{b}^*$: $\sigma^2\{\mathbf{b}^*\} = (\sigma^*)^2 r_{xx}^{-1}$

$(\sigma^*)^2$ = the error term variance for the transformed model

$(VIF)_k$ = the k th diagonal element of r_{xx}^{-1}

$$\Rightarrow \sigma^2\{b_k^*\} = (\sigma^*)^2 (VIF)_k = \frac{(\sigma^*)^2}{1 - R_k^2}$$

VIF for b_k^* : $(VIF)_k = (1 - R_k^2)^{-1}$, $k = 1, 2, \dots, p - 1$

R_k^2 : X_k is regressed on the $p - 2$ other X_k 's

- $R_k^2 = 0 \Rightarrow (VIF)_k = 1$: X_k is **not linearly** related to X_k 's
- $R_k^2 \neq 0 \Rightarrow (VIF)_k > 1$: indicate **inflated variance** for b_k^* as a result of the **intercorrelations among the X variables**

Variance Inflation Factor-VIF, cont'd

Perfect linear association with X_k s $\Rightarrow R_k^2 = 1 \Rightarrow VIF_k$ and $\sigma^2\{b_k^*\}$ are unbounded

Rule: largest VIF value among all X s $\Rightarrow$ as an indicator of the severity of multicollinearity

$$\max\{VIF_1, \dots, VIF_{p-1}\} > 10$$

Variance Inflation Factor-VIF, cont'd

Variable	b_k^*	$(VIF)_k$
X_1	4.2637	708.84
X_2	-2.9287	564.34
X_3	-1.5614	104.61

Maximum $(VIF)_k = 708.84$ $(\overline{VIF}) = 459.26$

Figure : VIF-Body Fat Example with three X s

- $VIF_3 = 105$
- $r_{13}^2 = 0.458^2 = 0.209764$, $r_{23}^2 = 0.085^2$: not large
- X_3 : $R_3^2 = 0.990$; strong related to X_1, X_2

Variance Inflation Factor-VIF, cont'd

Body Fat Example:

```
>summary(fit)
```

Call:

```
lm(formula = Y ~ X1 + X2 + X3)
```

Residuals:

Min	1Q	Median	3Q	Max
-3.7263	-1.6111	0.3923	1.4656	4.1277

Coefficients:

	Estimate	Std. Error	t value	Pr(> t)
(Intercept)	117.085	99.782	1.173	0.258
X1	4.334	3.016	1.437	0.170
X2	-2.857	2.582	-1.106	0.285
X3	-2.186	1.595	-1.370	0.190

Residual standard error: 2.48 on 16 degrees of freedom

Multiple R-squared: 0.8014, Adjusted R-squared: 0.7641

F-statistic: 21.52 on 3 and 16 DF, p-value: 7.343e-06

```
> anova(fit)
```

Analysis of Variance Table

Response: Y

	Df	Sum Sq	Mean Sq	F value	Pr(>F)
X1	1	352.27	352.27	57.2768	1.131e-06 ***
X2	1	33.17	33.17	5.3931	0.03373 *
X3	1	11.55	11.55	1.8773	0.18956
Residuals	16	98.40	6.15		

Signif. codes: 0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

```
> vif(fit)
```

X1	X2	X3
708.8429	564.3434	104.6060